DAPHNE AVKAROGULLARI

davkarog@andrew.cmu.edu | SF Bay Area, CA | linkedin.com/in/daphne-defne-avkarogullari | d-pha.github.io/portfolio | U.S. Citizen

EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, PA Expected May 2028
B.S. in Electrical & Computer Engineering; Minor: *Machine Learning*
GPA: 3.8 Dean's List with High Honors (Fall '24, Spring '25, Fall '25)
Relevant Coursework: Intro to Computer Systems, Principles of Imperative Computation, Structure & Design of Digital Systems, Intro to Machine Learning, NLP, Signals & Systems, Electronic Devices & Analog Circuits, Probability Theory, Linear Algebra

EXPERIENCE

Cisco — Software Engineer, Data/AI/Intelligent Systems Intern San Francisco Bay Area | Summer 2026

- Diagnosed a storage configuration issue causing excessive drive writes in a production data pipeline; the fix cut **drive writes ~50%**, easing sustained write load well below rated drive endurance and **roughly doubling projected SSD lifespan**. Required reasoning about concurrency/race conditions and **Python** benchmarking
- Designed a **reliability framework** (checksums/integrity checks) that reduced error volume in a system core to the policy pipeline; built with **Go, Python, YAML**, using **Kafka** and **Kubernetes**
- Supported **policy/configuration generation and distribution** infrastructure behind Cisco's cloud-delivered security service (**Umbrella**) and its Security Service Edge (**Secure Access**); gained hands-on experience with **AWS**

Amazon — Software Development Engineer Intern Seattle, WA | Summer 2025

- Prime Video Live Events. Launched **template selection & quick switching**, enabling operators to apply pre-tested settings to published live-sports events in one step; **projected to cut clicks by about 60% and task time by about 40%** (internal metrics)
- Delivered full-stack development across **Java (backend)** and **TypeScript/React (frontend)** with **DynamoDB**, feature flags, and safeguard mechanisms for safe rollout
- Shipped to production with a **unit/integration test suite (95–97% coverage)** and CI workflows to ensure reliability and prevent regressions

PROJECTS

Dunes Tower Defense ***(Build18 Officer's Choice Award; C++, OpenGL, OpenCV)*** Winter 2026

- Building a from-scratch tower-defense game (no engine) in **C++/OpenGL**; a **depth camera** reconstructs sand as a real-time heightmap, and **OpenCV** detects tower pieces to sync with game state and drive dynamic pathfinding

FanBuddy Robot ***(Build18; Embedded CV, OpenCV, microcontroller, CAD)*** Winter 2025

- **OpenCV-based** person-tracking fan with closed-loop control from CV outputs to motor commands; **about 100 ms** frame pipeline on Raspberry Pi; 3D-printed frame

Digital System Design: Datapath + FSM Hardware Unit (SystemVerilog, FPGA) Spring 2026

- Designed and implemented a **SystemVerilog** datapath and finite-state-machine-controlled hardware unit, synthesized and verified on FPGA; programmed a custom RISC-style ISA and gained working familiarity with **RISC-V** through coursework

15213/18213: Intro to Computer Systems (C, x86-64) Fall 2025

- Built a **multithreaded HTTP proxy** and a **dynamic memory allocator**; wrote a Unix shell and reverse-engineered x86-64 binaries via low-level exploits (GDB/objdump)

SKILLS

Languages: C, Java, Python, C++, Go, PHP, SQL, TypeScript/JavaScript, Assembly (x86-64), SystemVerilog, YAML
Frameworks, Tools & Platforms: React.js, OpenCV, NumPy, AWS (EKS, EC2, Secrets Manager, IAM, Certificate Manager), DynamoDB, RDS/Aurora, Kafka, Kubernetes, Ansible, Terraform, Jenkins, Docker, Git/GitHub, CI/CD, Linux, MATLAB
Concepts: Multithreading/concurrency, microservices, probability theory, linear algebra, RISC-V fundamentals; AI tooling (Claude Code, Codex, prompt engineering coursework)
Interests: Systems (OS/compilers/runtimes), hardware/software interface, ML/AI systems & efficiency, quantitative engineering

ACTIVITIES & HONORS

IEEE @ CMU | SWE @ CMU | Women in ECE @ CMU | Delta Gamma Booth (Electrical Committee & Buggy Team) | Carnegie Mellon Racing (software & electrical)
Build18 Officer's Choice Award (2026) | **Near Mitra Scholar** (2024) | **National Merit Finalist** (2023)